

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF ECONOMICS Inference in regression;
correlation and causality

Week 13 · Test for slope, two intervals and three traps · Lectures 1 and 2

Tashkent International University · Faculty of Economics

2025/2026 academic
year



One number, two questions

How did week 12 end

$\hat{\beta}_1 = 0.588033$ — Potato yield in Kazakhstan increased by an average of this many t/year each year.

But this is a sample estimate. If we take another nine years, it would give a different number. How different?

Structure of the week

Lecture 1 — the first question. A little new math: $SE(\hat{\beta}_1)$ and the t -test, both from weeks 9-10.

Lecture 2 — second question. There's not much new math, but the most difficult ideas of the entire course are here.

Today's two questions — and they are different

1. "How much do I trust $\hat{\beta}_1$?" $\Rightarrow$ standard error, t -test, confidence interval.
2. "Does $\hat{\beta}_1$ have causal significance?" $\Rightarrow$ this is a completely different question, and statistics alone cannot answer it.

A common mistake

Warning. After Lecture 1, you have a slope of $\hat{\beta}_1 < 0.001$, and it looks very reliable. Lecture 2 shows how easy it is to draw the wrong conclusion even in this case.

Small $\hat{\beta}_1$ — means "not a coincidence." It does not mean "causation."

Lecture 1 — How much confidence do we have in a line 1. $SE(\hat{\beta}_1)$ and the three things it depends on.

2. t -test and confidence intervals for slope.

3. Slope test = correlation test = ANOVA F .

4. Confidence Interval and Prediction Interval — the difference.

5. Testing with bootstrap and randomization.

6. How much error can the conclusion have if the condition is violated.

Lecture 2 — correlation and causation 7. Confounder: three conditions, two candidates.

8. Simpson's Paradox — two published real tables.

9. Selection Bias and Range Limitation.

10. Collider — A Case Where Stratification Hurts.

11. Why Causal Diagrams and Randomization Work.

Conditions CLO7 and CLO8. Infer and interpret slope correctly; explain the limits of causal inference from observational data.
Bloom: Analysis $\rightarrow$ Estimate.

Conclusion for the slope

SE(β_1) — and the third factor is often forgotten

Standard error (F184)

$$SE(\beta_1) = \frac{s}{\sqrt{S_{xx}}}, \quad S_{xx} = \sum (x_i - \bar{x})^2$$

Three ways to make it smaller: 1. s small — the model fits well. 2. n large — it grows with $\sqrt{n}$. 3. x scattered — and this is almost always forgotten.

In our model

$s = 0.667950$, $n = 60$ $SE(\beta_1) = 0.667950/\sqrt{60} = 0.086232$
 $SE(\beta_0) = 0.410547$ (F187)

We measure the third factor: the same n , the same s

n values ($n = 9$) SE ratio

n values	SE	Ratio	Notes
0, 1, ..., 8 (ours)	60.00	1.0000	only 3-5 (compressed)
4.00	3.8730		only 0 and 8 (edges)
128.00	0.6847		

Compressing n increases the standard error by 3.87 times. But rounding to the edges reduces it to 0.68 — with the same n , the same s .

A common mistake

This is a design issue, not a calculation issue: you choose where to take n when you design the experiment.

But there are two limitations. (a) Expanding n leads to extrapolation (week 12). (b) If you only take the edges, you can't check for linearity — there will always be a straight line between the two points.

Test, interval — and three tests are one question

Slope test (F185-F186)

$$\beta_1 = \frac{\hat{\beta}_1 - \beta_1(0)}{SE(\hat{\beta}_1)}, df = n - 2$$

For $\beta_1 = 0$: $\hat{\beta}_1 = 0.588033$
 $t = 6.819199$

$p = 0.00024883 \Rightarrow$ rejected.

$0.975(7) = 2.364624$ 95 % CI:
[0.384127; 0.791940] — zero outside.

Conditions Interpretation: yield increased by an average of 0.5880 t/ha per year; accuracy ± 0.2039 t/ha. Always include the interval — a rule that has been in effect since week 9.

Three tests, one question

statistics

Slope $t = 6.81919887$ 0.00024883 Correlation
(F188) $t = 6.81919887$ 0.00024883 ANOVA
(F189) $t = 46.50147329$ 0.00024883

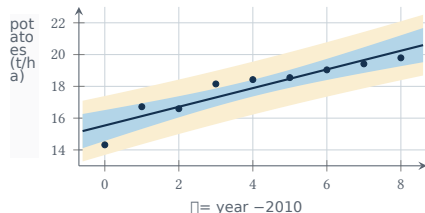
$t^2 = 46.50147300$; difference from t is 0.00. The difference between the slope and the correlation t is 0.00 (the same number).

“Is the slope zero?” and “Is the correlation zero?” are the same question, written in three different languages.

A common mistake

Note: This is only true for models with a single explanatory variable. In multivariate regression, t tests the entire model, and T tests a single coefficient — and they can easily give conflicting results.

Two intervals: where is the line and where is the next observation



Conditions Inside (blue) — 95% confidence interval for the mean response; outside (yellow) — prediction interval. Both widen as they move away from $\bar{x}$: hourglass shape.

Difference in one term (F190 and F192)

$$SE_{\text{mean}} = \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}}$$

$$SE_{\text{predict}} = \sqrt{1 + \frac{(x^* - \bar{x})^2}{S_{xx}}}$$

Addendum 1 — the new observation's own randomness; knowing the line does not eliminate it.

At $x^* = \bar{x}$, $\bar{x} = 9$

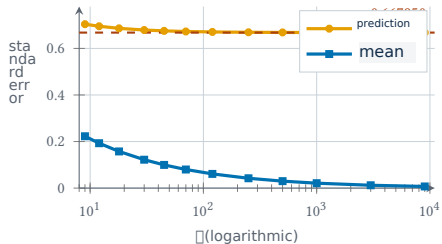
$SE_{\text{mean}} = 0.222650$ $SE_{\text{predict}} = 0.704082$ Ratio = 3.162278 times.

At $x^* = 0$ (outlier): SEs are 0.410547 and 0.784000; IO [14.5665; 16.5081], BO [13.6834; 17.3913].

A common mistake

The most common mistake is to give the mean interval when predicting the yield of a single year: it is about three times narrower, and therefore seems three times more reliable.

□□: one tends to zero, the other does not



	SEmean	SEforecast
9	0,222650	0,704082
45	0,099572	0,675331
900	0,022265	0,668321
9000	0,007041	0,667988

This week's most important sentence may be

With infinite data, we know the line perfectly — $SE_{mean} \rightarrow 0$.

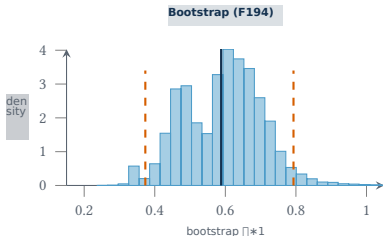
But we still can't say for sure what next year's crop will be: $SE_{forecast} \rightarrow \square\square = 0.667950$ and won't fall below it.

Why is this not a flaw in the model

□□ — not a flaw in the model, but a coincidence of the world. Weather, markets, politics — they are not in $\square$ and no sample size can eliminate them.

"The more data we have, the more accurate our predictions" is partly true: the line is defined, but the prediction is almost completely undefined.

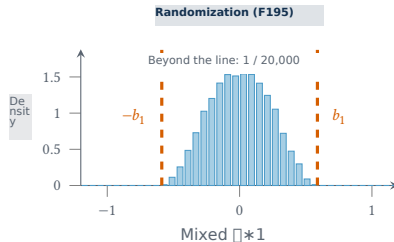
Two independent tests: bootstrap and randomization



Conditions 20,000 replicates, seed 20261301. Bootstrap SE = 0.112797 vs formula 0.086232 (ratio 1.3081). Percentile IO [0.373266; 0.793133] · β -IO [0.384127; 0.791940].

Audience question

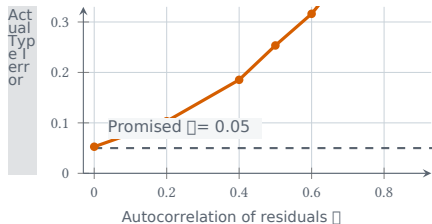
Here the bootstrap is not an alternative to the β -interval — $\beta = 9$ is too small and the bootstrap SE formula is 1.31 times larger than the SE. It is a checker: if three completely different paths lead to the same conclusion, the conclusion does not follow from the choice of formula.



Conditions

$\beta = 0$: β and β are not related $\Rightarrow \beta$ can be mixed. Cases where $|\beta| * 1| \geq 0.588033$: 1/20,000, $\beta = 0.000050$. β -test $\beta = 0.000249$ — same conclusion.

How much error will the conclusion have if the condition is violated — let's measure



Conditions α_0 true (true slope zero), $n = 27$, AR(1) residuals, 20,000 repetitions. $\alpha = 0.0527$; $\alpha = 0.9$ 0.6207 — that is, a false positive in most cases.

Uzbek wheat row at week 12

Residuals lag-1 = 0.789704 (boundary 0.377195) — the condition is violated.

Effective sample size (F199): $n_{\text{eff}} = n(1 - \alpha)$

$$= 27 \cdot \frac{0.210296}{1.789704} =$$

3.172590 27 years is approximately equal to 3.17 independent observations.

SE correction coefficient $\sqrt{(1 + \alpha)/(1 - \alpha)} = 2.917258$. Simple SE = 0.012480 $\rightarrow$ corrected 0.036408.

Has the conclusion changed?

Simple: $\alpha = 10.806238$, $\alpha = 6.556 \times 10^{-11} \Rightarrow$ reject. Corrected: $\alpha = 3.704245$, df ≈ 1.1726 , $\alpha = 0.138445 \Rightarrow$ not rejected.

In this example, the conclusion has CHANGED. The violated condition does not violate α_1 , but the belief in it — and here it has been completely destroyed.

Correlation and Causation

There is a connection. Now what?

If x and y are related, there are at least five explanations (F201)

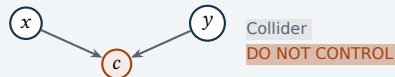
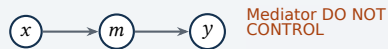
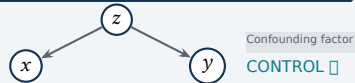
1. $x \rightarrow y$ — the answer we need 2. $y \rightarrow x$ — reverse causality 3. $x \leftarrow z \rightarrow y$ — a confounding factor 4. selection bias — who we are looking at 5. coincidence

The p -value only loses its FIFTH

$p = 0.00024883$ — this means “not a coincidence.”

The other three explanations remain the same, no matter how small p is. If you increase p , p gets smaller and the confounding factor stays there.

Three structures, one view



Audience question

In the data, all three look the same: the third variable is associated with both x and y . Only domain knowledge can tell which one it is—never the data itself.

Confounding factor: two plausible candidates, two different results

Question and raw answer

Do PG-13 and R films differ in profitability? ("Profitable" = worldwide gross > 2.5× budget.)

	rating	gross	profit share
PG-13	358 892	0.401345	R 409 1097 0.372835

Raw difference = 0.028510 (in favor of PG-13).

CANDIDATE 1: budget — plausible, but NOT

Average budget: PG-13 \$43.78 million, R \$23.85 million (83.58% difference).

✓ budget ~rating: $\beta = 0.280993$, $\sigma^2 = 2.06 \times 10^{-37}$

✗ budget ~utility: $\beta = 0.041462$, $\sigma^2 = 0.0645$ — not significant

The second condition is not met $\Rightarrow$ budget is not a confounding factor. Anyway, if we look at it in layers, the difference is 0.028510 $\rightarrow$ 0.034713 — it changed little, as expected.

CANDIDATE 2: genre — YES

✓ genre ~rating: $\beta^2 = 85.3574$, $\sigma^2 = 1.27 \times 10^{-17}$

✓ genre ~utility: $\beta^2 = 41.2280$, $\sigma^2 = 2.41 \times 10^{-8}$

genre is not a consequence of the rating (genre is in the script, rating when the film is ready)

genre	PG-13	R	difference
Action	0.4181	0.4012	+0.0169
Adventure	0.4194	0.2857	+0.1336
Comedy	0.3849	0.3819	+0.0030
Drama	0.3620	0.3067	+0.0553
Horror	0.6667	0.5395	+0.1272

Weighted (F197) difference = 0.044433 vs raw 0.028510 $\Rightarrow$ +55.85%.

Unexpected direction — and it's real

Usually, the correction makes the difference smaller. Here it is: R movies are more useful in low genres (Drama), and PG-13 in high genres. The genre was masking the difference — negative bias.

Simpson's paradox — first: is it in our data?

Definition (F198)

If the general conclusion is reversed at all levels, then this is Simpson's paradox.

The most extreme manifestation of a confounding factor: the correction does not reduce the result, but inverts it.

And this is also the result

Simpson's paradox is possible, but not mandatory. Because textbooks show it so often, it seems like a phenomenon that occurs at every step - in fact, it is not.

The correct conclusion is: “you can't tell without stratification” — not “it's always reversed.”

Systematic search

We checked the triples in both real data sets (□, □, group):

Checked triples: 104 Complete
reversals found: 0

There is no Simpson's paradox in our data.

Therefore, below is the PUBLISHED real table

Both are valid, both with source:

1. Berkeley University, 1973 — graduate admissions. Bickel, Hammel, O'Connell, Science 187(4175), 398-404 (1975). Partial reversal and subtle interpretation.

2. Kidney stone treatment study, 1986. Charig et al., BMJ 292(6524), 879-882. Complete reversal.

Berkeley 1973: the most famous real example

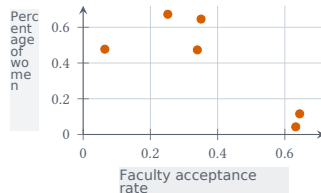
Fac. Male Female who arz. percent arz.
percentage higher

A 825 62.06 % 108 82.41 % FEMALE B 560 63.04 % 25
68.00 % FEMALE C 325 36.92 % 593 34.06 % male D
417 33.09 % 375 34.93 % FEMALE E 191 27.75 % 393
23.92 % male F 373 5.90 % 341 7.04 % FEMALE

TOTAL 2691 44.52 % 1835 30.35 % MALE

Common mistake

Women are higher in four of the six faculties; the overall result is 14.1645 percentage points in favor of men.



Reason

Correlation (acceptance rate, percentage of women) = -0.785794 : Women applied more to faculties with lower acceptance rates. Faculty is a confounding factor: it affects both who applies where and the acceptance rate.

Important caveat

This does not mean that there is “no discrimination”: the total number does not indicate discrimination at the faculty level, only that. Why women apply to those faculties is a separate question, and it is not in the data.

Kidney stone, 1986: complete reversal

	A (open)	B (percutaneous)	ct/total	percentage	ct/total	percentage
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Small stone	81/87	93.1 %	234/270	86.7 %
Large stone	192/263	73.0 %	55/80	68.8 %

TOTAL	273/350	78.0 %	289/350	82.6 %
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Theorem

A is better in both groups (+6.4 and +4.3 percentage points); B is better overall (+4.6). Neither number is wrong — both are correct.

Reason: Stone size is a confounding factor

Method A was used for large stones in 75.14% of cases. Method B was used for small stones in 77.14% of cases.

Large stones are generally more difficult: success rate 72.01% vs. 88.24% for small stones.

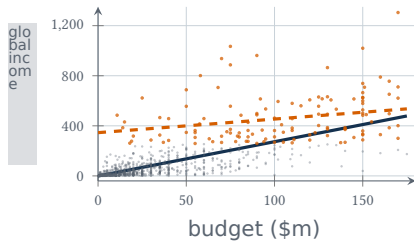
Doctors chose A in severe cases — so A looks “bad.”

There's a name for this trap

Confounding by indication is the most common trap in medicine.

Which answer is correct? The answer is a composite: the size of the stone determines which method a patient needs. The total number is not anyone's.

Selection bias: looking only at the winners



Terms Red — top 10% by income. Dashed line — regression adjusted for them only.

selection $\square \square 1 \square$

all movies	2436	2.739510	0.700195
top 50%	1218	2.408297	0.591148
top 25%	609	1.942877	0.468804
top 10%	244	1.085625	0.246692
top 5%	122	0.240133	0.055240

No reason has changed

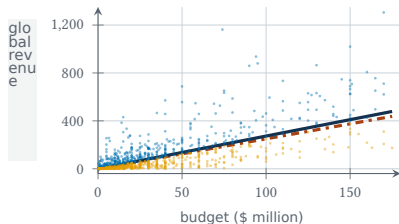
Slope 2.7395 $\rightarrow$ 1.0856 (0.3963 times), correlation 0.7002 $\rightarrow$ 0.2467.

Only who we watch has changed. This is a range restriction.

Audience Question

“We studied the most successful companies and found a common characteristic”—this is exactly the kind of research design that makes this mistake. The unsuccessful ones may have the same characteristic; they are not in the sample.

Collider: case where layering is HARMFUL



Conditions Red dashed line — definition of “profitable”: revenue = 2.5 × budget. Blue — profitable, yellow — unprofitable.

“Profitable” is the result of BOTH variables

profitable = [revenue > 2.5 × budget] — depends on both budget and revenue. So it is a collider.

group 0 1

all	2436	2.739510	0.700195	useful
959	3.875674	0.824915	useless	
1477	1.538807	0.864880		

The correlation in both parts is stronger than the total (0.8249 and 0.8649 vs 0.7002) — this is a signature of the collider.

Why — the geometry is visible

The dividing line is oblique. Each half of the cloud cut by the oblique line extends parallel to that line — so the coupling in both halves is enhanced.

Audience question

Rule: layering is necessary for the distractor; harmful for the collider. And they look the same in the data.

Why randomization works — let's measure it

Experiment (hypothetical world, we know the truth)

$X \rightarrow Y$ and $X \rightarrow Z$, but the effect of $X \rightarrow Y$ is exactly zero.

$n = 200$, 20,000 repetitions, seed 20261500.

Two cases: (a) observation — Y naturally depends on X ; (b) experiment — X is assigned randomly.

Theorem

method average estimate $|\text{estimate}| > 0.1$

observation	0.640649	1.000000	random
experiment	0.000130	0.155800	

True effect = 0. An observational study will always give a nonzero answer around 0.6406.

Why experiments work

Randomization cuts off Z from X . The $X \rightarrow Z$ axis disappears, so the $Z \leftarrow X \rightarrow Y$ path disappears.

And most importantly: this works even if we don't measure Z . Stratification only corrects for the factors you know and measure; randomization corrects for everything.

But experiments don't always work

Productivity can't be assigned randomly. Harmful effects on humans cannot be randomised.

That is why observational research is needed — and why this week's whole point is to know the pitfalls.

Prerequisites It is possible to draw causal inferences from observational data, but this requires additional hypotheses, and these come from the field, not from the data. (Weeks 14-15 and subsequent courses.)

**Summar
y**

Common Student Mistakes and Misconceptions

1-5 1. " $p < 0.001 \Rightarrow \text{cause.}$ " p only misses the chance explanation — one in five.

2. Instead of the prediction interval, the mean interval is given. In our example, it is 3.16 times narrower.

3. "More information $\Rightarrow$ more accurate prediction." They define a line; the prediction stops at $\hat{y}$.

4. Thinking that $SE(\hat{\beta}_1)$ depends only on $\hat{\beta}_1$. The dispersion of $\hat{\beta}_1$ is just as important.

5. Using the simple SE even when the condition is violated. $\hat{\beta}_1 = 0.79$, $\hat{\beta}_0 = 27$ is actually $\beta_{\text{eff}} = 3.17$.

6-10 6. "Simpson's paradox always happens." We checked 104 triples and found none.

7. "Correct" any third variable. Budget was not a confounding factor; fixing the collider would hurt.

8. "Correcting always reduces the effect." Genre increased it by 55.85%.

9. Study only the successful. In the top 10%, $\hat{\beta}_1$ fell from 0.70 to 0.25.

10. "The Berkeley data proves there is no discrimination." It only shows where the total number came from.

Summary: Three Sentences

What to Report

1 β_1 units with 2 $SE(\beta_1)$ and 95% confidence interval 3 β_1 , df and 4 β_2 and β_3 (week 12) 5 Which L-I-N-E condition was tested and the result 6 The prediction — stating which interval it is 7 What confounding factors were taken into account 8 What was not proven and why

In Three Sentences

1. $SE(\beta_1)$ depends on three things, and the third — the dispersion of β_1 — is a design issue, not a calculation issue.

2. Knowing the line and knowing the next observation are two different things, and the difference between them never disappears.

3. Small β_1 rejects the explanation, but only if it is. To eliminate the other four, you need a design — and if the design is observational, you have to say what you did not prove.

Next step

Week 14: Data project lab — complete analysis from question to conclusion, recursive report, and AI use statement.

Project 12 is the product of this week: what was not proven and why.

References

Main reading

►Çetinkaya-Rundel M., Hardin J. Introduction to Modern Statistics (IMS, 2nd ed.), chapters 24–26 (inference in regression). Open source: openintro.org/book/ims ►Newbold P., Carlson W., Thorne B. Statistics for Business and Economics (9th ed.), 11.4–11.8.

This week's required reading — both short

►Bickel P., Hammel E., O'Connell J. (1975). Sex bias in graduate admissions: data from Berkeley. *Science* 187(4175), 398–404. — 7 pages; original of today's table.
►Charig C., Webb D., Payne S., Wickham J. (1986). Comparison of treatment of renal calculi... *British Medical Journal* 292(6524), 879–882. — 4 pages.

Supplement

►Simpson E. H. (1951). The interpretation of interaction in contingency tables. *JRSS B* 13(2), 238–241. ►Hernán M., Robins J. Causal Inference: What If. — open access; Chapter 6 (causal diagrams) relevant to this week. ►Cunningham S. Causal Inference: The Mixtape. — open access; Chapter 3 (directional graphs), with collider examples. ►Rosenbaum P. (2017). Observation and Experiment. — an introduction to the logic of observational research.